Optimization for Deep Learning (OPT4DL)

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High Confidence Guarantees

Average vs High Confidence



- Recall we showed for SGD

$$\mathbb{E}[\text{error}] \leq \text{Bound}(T) \tag{1}$$

  This means if running SGD many many times, on average it perform well

- Not satisfactory, we want good performance after only one-time training. That is, given $0 < \delta < 1$ we want with probability exceeding $1 - \delta$

$$\text{error} \leq \text{Bound}(T, \delta) \tag{2}$$

  or equivalently

$$\Pr\{\text{error} \geq \text{Bound}(T)\} \leq \delta \tag{3}$$

- Can SGD deliver such results?

Stronger Results under Stronger Assumptions



- Recall our SFO assumption (written slightly differently here)

$$g_t = \nabla f(w_t) + e_t, \quad \mathbb{E}[e_t|w_t] = 0, \quad \mathbb{E}[\|e_t\|^2|w_t] \leq \sigma \tag{4}$$

- We can assume the noise is light-tailed (or sub-Gaussian)

$$\mathbb{E}[\exp(\|e_t\|^2/\sigma^2)|w_t] \leq 1 \tag{5}$$

  Example: Gaussian noise, uniformly bounded noise, etc.

- Combined with concentration of measure results (deviation from the mean), e.g., Azuma–Hoeffding, we can show our previous bounds hold with high confidence:

$$\text{error} \leq \text{Bound}(T, \log \frac{1}{\delta}) \tag{6}$$



Martingale Difference Sequence (MDS)

A sequence of random variables $X_1, \ldots, X_T$ is an MDS if $\mathbb{E}[X_t | z_1, \ldots z_{t-1}] = 0$ where $z_1, \ldots, z_T$ is another sequence of random variables

- Think of it as a generalization of zero-mean independent random variables.
- In the context of DL optimization, very often

$$X_t = g_t - \nabla f(w_t), \qquad \mathbb{E}_{z_t}[g_t - \nabla f(w_t) | z_1, \ldots z_{t-1}] = 0 \tag{7}$$

  the term we dealt with in the proofs for SGD before.
- When assuming bounded gradient or light-tailed noise, we can show $\mathcal{O}(\sqrt{T \log \frac{1}{\delta}})$ growth of noisy terms

$$\sum_{t=1}^{T} \langle g_t - \nabla f(w_t), \nabla f(w_t) \rangle, \qquad \sum_{t=1}^{T} \eta \| g_t - \nabla f(w_t) \|^2 \tag{8}$$

Stronger Results under Stronger Assumptions (Cont'd)



- Could be violated in practice. A remedy is clipping or normalization:

$$g_t = \nabla f(w_t) + e_t, \quad w_{t+1} = w_t - \eta \min\{\max\{-C, g_t\}, C\} \quad \text{or} \quad w_{t+1} = w_t - \eta \frac{g_t}{\|g_t\| + \kappa} \tag{9}$$

  for some clipping parameter $C > 0$ and a small constant $\kappa > 0$ for numerical stability
- These will control the noise and could deliver high confidence guarantees
- Connection to advanced training methods (adaptivity) and other contexts (privacy).
- Appear in diverse learning settings, e.g., the PPO method in reinforcement Learning (RL)
- Without such safeguarding mechanisms, the growth of the accumulation of noisy terms could be larger w.r.t. to δ.



Markov Inequality

For any positive random variable X and constant $\lambda > 0$ we have

$$\Pr\{X > \lambda\} \leq \frac{\mathbb{E}[X]}{\lambda} \tag{10}$$

- Let $X = \text{error}$ and $\lambda = \text{Bound}(T)/\delta$
- By Markov

$$\Pr\{\text{error} > \text{Bound}(T)/\delta\} \leq \delta \frac{\mathbb{E}[\text{error}]}{\text{Bound}(T)} \leq \delta \tag{11}$$

or equivalently, with probability at least $1 - \delta$,

$$\text{error} \leq \text{Bound}(T)/\delta \tag{12}$$

- Terrible dependence on confidence δ (i.e., $1/\delta$ vs our desired $\log 1/\delta$)



- We will boost SGD by running multiple SGDs in parallel and performing post-optimization
  - number of parallel runs S, number of samples for post-optimization N
  - Run S versions of SGDs in parallel and call their outputs $\hat{w}_s$, for $s = 1, \ldots, S$
  - Choose the best output $\hat{w}$:

$$\hat{w} = \arg \min_{w \in \{\hat{w}_1, \ldots, \hat{w}_S\}} \|G(w)\|, \qquad G(w) = \frac{1}{N} \sum_{j=1}^{N} \nabla \ell(w, z_j^{po}) \tag{13}$$

    or use function values instead

- Total Complexity: Recall for SGD total oracle/gradient complexity is $\mathcal{O}(BT)$ and $\mathcal{O}(T)$ when B is some constant. For SGD-PO:

$$\text{Oracle complexity } = S \times T + S \times N = (N + T) \times S \tag{14}$$



Theorem

If

$$N = \Omega(\frac{\sigma^2 \log \frac{1}{\delta}}{\epsilon\delta}), \quad T = \Omega(\frac{1}{\epsilon^2}), \quad S = \Omega(\log \frac{1}{\delta}) \tag{15}$$

SGD with Post-Optimization can find a solution $\hat{w}$ such that with probability at least $1 - \delta$ it holds $\|\nabla f(\hat{w})\|^2 \leq \epsilon$.

- Thus, the total complexity in terms of ϵ and δ is

$$\frac{\log^2 \frac{1}{\delta}}{\epsilon\delta} + \frac{\log \frac{1}{\delta}}{\epsilon^2} \tag{16}$$

- Proof relies on showing the ensembling/post-optimization enables boosting the simple Markov concentration result, hence improving the dependence on δ.